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Technical Design Document: The Human-Centricity Gap in AI-driven HR systems

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1. Problem Definition

AI is increasingly used in HR, covering everything from hiring to performance reviews to workforce planning. On paper, this looks like progress. In practice, these systems have a serious problem: they were built to make HR more efficient, not to actually support the people inside organizations. That gap is what this project addresses.

The first issue is that most HR AI systems do not explain themselves. When an employee is passed over for promotion or flagged in a performance review, they rarely get a clear reason. That can be quite frustrating and even detrimental to the person. People need to feel that decisions affecting their careers are fair and understandable. When they do not, trust breaks down, and people start looking for the exit. This is also becoming a legal issue. The EU AI Act now requires that high-risk AI systems, which include most consequential HR tools, be transparent and auditable. Many existing systems simply were not built with that in mind.

The second issue is bias. AI systems learn from historical data, and historical data reflects historical inequalities. If a company has historically underpromoted women or underrepresented certain groups, an AI trained on that data will likely continue those patterns unless something actively intervenes. Most current systems have no meaningful mechanism for catching this. The result is that automated decisions quietly reproduce structural inequity, not because anyone intended it, but because no one built in the safeguards to stop it.

The third issue is that these tools are blind to burnout. HR analytics platforms are good at tracking output, task completion, and attendance. What they cannot track is how someone is actually doing. An employee burning out will often keep hitting their targets right up until they quit or collapse. The warning signs are in the qualitative stuff, such as how they write in team channels, how they respond in surveys, and whether their language has shifted over time. Current systems ignore all of this because it is harder to measure. That is exactly where generative AI can do something genuinely useful.

The fourth issue is surveillance. To detect well-being problems early, a system needs fairly detailed data on how employees work. But the more detailed that monitoring is, the more it starts to feel like being watched. Employees who feel watched are less likely to speak up, feel less psychologically safe, and are more likely to disengage. This is a real tension that cannot be solved just by writing a good privacy policy. It has to be solved at the design level, by giving employees genuine visibility and control over their own data.

These are not minor technical issues to see to at a later stage. They reflect how HR AI has been designed from the start, centered on what organizations want to optimize rather than what employees actually need. This project proposes a system built the other way around, one where employee wellbeing is the starting point, and everything else, the explainability, the bias detection, the qualitative analysis, is designed to serve that goal.

2. Target Users

This system has four users. Each one sees a different version of the system depending on their role, and access is strictly controlled. That boundary is not a policy decision; it is enforced at the architecture level through role-based access control, meaning the system itself determines what each user can and cannot see.

2.1 The People Partner

The People Partner is the primary user. They are HR professionals responsible for workforce health and retention, and they are the ones who translate system outputs into real organisational decisions.

Most HR intervention right now is reactive. Someone burns out or resigns and the organisation responds after the fact. This system changes that by giving the People Partner early warning signals generated from patterns in workforce data that would not be visible through manual analysis alone.

Technically, they upload workforce data in CSV or JSON format, which the system anonymizes and processes through a burnout risk classification model. They configure the thresholds that determine when the system raises an alert, and they interact with explainable AI outputs that show not just a risk score but the specific factors behind it. That explainability layer is what makes the system defensible, both internally to employees and externally under frameworks like the EU AI Act.

They have the highest access level in the system but individual employees remain anonymised unless a formally triggered escalation protocol requires otherwise.

2.2 The Team Lead

The Team Lead is a line manager handling day-to-day team performance. They are often the first person to sense when something is wrong within a team; however, without structured data to support that instinct, it rarely translates into meaningful action.

The system gives them a team-level dashboard that visualizes workload distribution, participation patterns, and early indicators of strain or disengagement across the team. When the system detects something worth acting on, it generates a plain language prompt suggesting a specific response, for example, a check in conversation or a workload rebalancing. The system also surfaces patterns in management behaviour that may be contributing to team stress, something most HR tools do not account for at all.

Individual employee data is completely off limits to this user. No names, no individual risk scores, no information that could identify a specific person. That restriction is not a setting that can be adjusted; it is a hard boundary in the system architecture enforced through role-based access control.

This user sits at the intersection of two important labour rights principles, the right to disconnect and the right to fair working conditions. The system is designed to support both rather than give managers additional tools to intensify monitoring.

2.3 The Employee

The employee is for whom the system is ultimately built. Their data powers the system, but they are not a passive data source. They are active participants with meaningful control over how their information is collected, processed, and acted upon.

The central risk in any workplace wellbeing system is that employees experience it as surveillance repackaged as support. This system addresses that directly through a personal transparency interface where every employee can see exactly what data the system holds on them, what it has inferred from that data, and what it is recommending as a result. Nothing about them is hidden from them.

Their identity is never visible to their manager or team lead. This is not a policy promise; it is a structural feature of how the data pipeline is built. Anonymisation happens at the point of ingestion, before data is processed or stored in a form accessible to other users. Employees can also challenge system outputs they believe are not inherently true, and that feedback is routed directly into the bias monitoring layer to inform ongoing model evaluation.

This consent-first architecture is both the ethical foundation of the system and a practical necessity. A system that employees do not trust will produce data that does not reflect reality, which makes every output unreliable. Trust is not a feature added at the end; it is a core design requirement.

2.4 The Compliance and Legal Officer

The Compliance and Legal Officer does not interact with the system daily. Their role is to verify that the system is doing what it claims, consistently, fairly, and in line with current regulation, most critically the EU AI Act, which classifies consequential HR AI tools as high-risk systems subject to strict transparency and auditability requirements.

They have access to full system audit logs, bias detection reports, anonymisation verification records, and a history of all automated decision outputs. When the system produces patterns that suggest a protected group is being disproportionately flagged as high risk, the Compliance Officer is responsible for determining whether that reflects a genuine bias in the model, an issue in the training data, or a legitimate signal in the workforce data itself. That distinction matters legally and technically, and it requires someone with both regulatory knowledge and access to the system's internal outputs.

They are also the formal escalation point for any employee who wants to challenge a system output or raise a concern about how their data is being used. That channel exists because the EU AI Act requires affected individuals to have a meaningful route to contest automated decisions, and this role is how the system delivers that in practice.

Their access covers system behaviour and audit data comprehensively, but does not extend to individually identifiable employee records. The distinction is important: they are auditing the system's integrity, not investigating the people inside it.

3. System Architecture

This system is built as a single pipeline that takes in raw workforce data, processes it through an LLM, and delivers role-specific outputs to each user. The entire prototype is built in Python, developed in Google Colab, and presented through a Streamlit interface. Three things are non-negotiable in how it is designed: every AI output must be explainable, individual employee data must never reach a manager, and every action the system takes must be logged and auditable.

3.1 How it works at a high level

A People Partner uploads a workforce dataset through the Streamlit interface. The moment that file enters the system it is anonymised and prepared for processing. The cleaned data, both the numerical workforce metrics and any free text survey responses, is passed to an LLM through a structured prompt. The LLM analyses the quantitative patterns, reads the sentiment in the qualitative text, identifies burnout risk signals across both, and generates a plain language explanation of what it found. The output is then filtered by role and delivered to the relevant user. Every step of this is logged automatically for audit purposes.

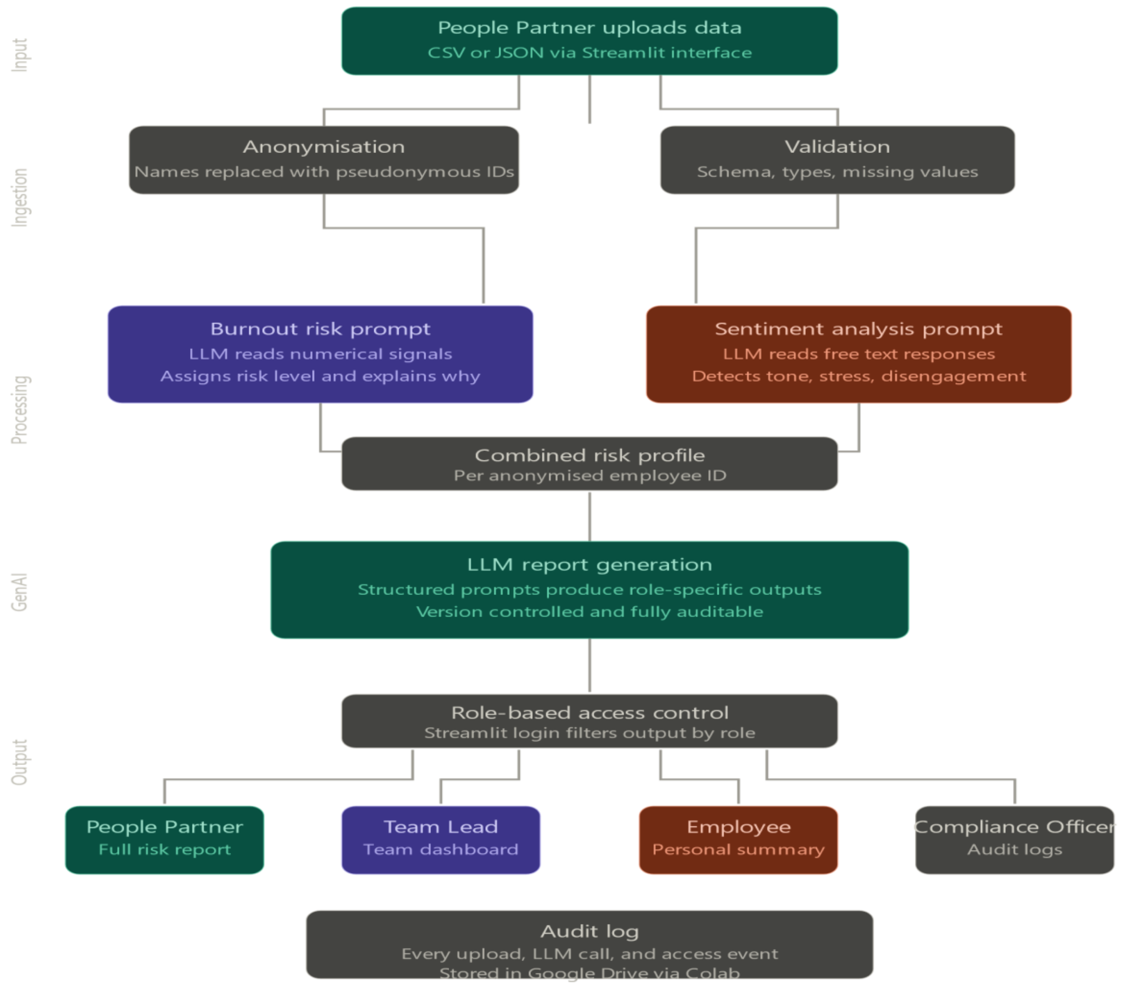

Figure 1: *System Architecture Design.*

3.2 Data input layer

The system accepts workforce data uploaded as a CSV or JSON file through the Streamlit interface. As soon as the file lands, two things happen before the LLM sees any of it.

First and most importantly, anonymisation. Employee names and any directly identifying information are stripped and replaced with pseudonymous IDs. The table that maps those IDs back to real identities is stored separately and is only accessible to the system administrator. Everything downstream in the pipeline works entirely with anonymised data and never processes a real name.

Second, validation. The system checks that the uploaded file has the correct structure, meaning the right columns are present, data types are as expected, and there are no missing values in fields the system depends on. If something is wrong, the system returns a specific error message telling the People Partner exactly what needs fixing before they resubmit.

3.3 Processing layer

This is where the LLM does its two core jobs.

The first is burnout risk analysis. The anonymised numerical data, covering signals like overtime frequency, leave patterns, output consistency, and participation rates, is passed to the LLM with a structured prompt that instructs it to identify patterns associated with burnout risk and assign a risk level to each employee ID. The prompt also instructs the LLM to explain which specific signals drove each assessment, which is what makes the output explainable rather than just a label with no context.

The second is sentiment analysis. Any free-text inputs, such as survey responses, are passed to the same LLM through a separate prompt that instructs it to assess emotional tone, identify stress indicators, and flag signs of disengagement in the language used. The output is a sentiment profile per anonymised employee ID that sits alongside their quantitative risk assessment.

Both outputs are combined into a single risk profile per employee that moves into the next layer.

Using one LLM for both tasks is a deliberate design choice. LLMs are deep learning models built on transformer architecture, trained on vast datasets to understand both numerical context and natural language. That dual capability makes them well-suited to a system that needs to reason across structured data and free text simultaneously. A single model also simplifies the pipeline, reduces points of failure, and makes the system easier to maintain and explain.

3.4 Generative AI layer

Once the risk profiles are produced, the LLM is called again to generate the final user-facing outputs. This is the generative AI layer of the system, and it is where the raw analysis becomes something a non-technical user can actually act on.

For the People Partner, the LLM generates a narrative report that explains which patterns were detected across the workforce, what is driving individual risk scores, and the recommended next steps. For the Team Lead, the same underlying data produces an aggregated team-level summary with no individual detail included in the prompt. For the employee, it generates a personal wellbeing summary written in plain, approachable language that explains what the system found and what it suggests.

Every prompt is structured and version-controlled. That means every output can be traced back to the exact prompt and the exact data that produced it. This is a direct requirement of the EU AI Act, which mandates that automated decisions affecting people must be explainable and contestable by the individuals they affect.

The LLM component is built to be modular. Whether the final implementation uses the Claude API, OpenAI, or another provider, the prompt structure and output format stay consistent. Only the API call changes.

3.5 Output and access control layer

Once outputs are generated, role-based access control determines what each user actually sees. This is enforced in the Streamlit interface through a login system that assigns each user to a role on sign in.

The People Partner sees anonymised individual risk profiles, team level summaries, and the full LLM generated report. The Team Lead sees only the aggregated team dashboard with no individual scores or identifiers anywhere on their view. The employee sees only their own personal summary and their transparency interface showing exactly what data the system holds on them. The Compliance Officer sees audit logs, bias detection reports, and system-level outputs, but no individual employee records.

These permissions are not settings that can be adjusted through the interface. They are hardcoded into the application logic, which means changing them requires a direct change to the codebase and creates an automatic record of that change.

3.6 Storage and audit layer

All data is stored in a structured format with separate tables for anonymised input data, LLM outputs, generated reports, and audit logs. The mapping table that links pseudonymous IDs back to real identities sits in complete isolation, accessible only under administrator credentials.

Every action in the system is logged. Every file upload, every LLM call, every time a user accesses a report. In the prototype, this is implemented using Google Drive as persistent storage connected to the Colab environment, with logs written to a structured format that can be exported and reviewed at any time. Those logs are what the Compliance Officer works with and what would be produced in the event of a regulatory review under the EU AI Act.

3.7 Known limitations

This is a working prototype, and it has real constraints worth being upfront about.

The system uses synthetic data generated to reflect realistic HR patterns, because real employee data is not accessible at this stage. The LLM is prompted and evaluated against that synthetic dataset, and results should be interpreted in that context. Transitioning to real organisational data would require a dedicated bias audit before any deployment.

LLMs carry a hallucination risk, meaning the model can produce plausible-sounding outputs that do not accurately reflect the underlying data. This is reduced by keeping prompts tightly structured and grounding every output directly in the actual data values before the LLM generates its summary. Human review of any LLM output before action is taken is a design requirement, not a suggested step.

LLMs are also less precise on structured numerical data than dedicated classification models. This is a known trade-off of using a single model across both tasks and is acknowledged as a limitation of this prototype. Future iterations of the system could introduce a dedicated classification model for the quantitative stream while retaining the LLM for explanation generation and qualitative analysis.

Google Colab has session-based compute limits, meaning the prototype is not built for continuous or production-scale use. This is a constraint of the development environment and is acknowledged as a boundary of the prototype rather than the proposed system design.

4. AI Model Selection

The AI model selected for this system had to meet one core requirement. It needed to reason across both structured numerical data and unstructured natural language in a single pipeline and produce outputs that a non-technical user could actually understand and act on. After evaluating the available options against that requirement, Gemini 2.5 Flash by Google DeepMind was selected as the primary model for this prototype.

4.1 Why a single LLM rather than multiple models

An earlier consideration in the design process was whether to use separate specialised models for each task, one for burnout risk scoring on the numerical data and a separate one for sentiment analysis on the survey text. That approach was not plausible for two reasons.

First, it would introduce unnecessary complexity into a prototype system. Multiple models mean multiple points of failure, multiple APIs to manage, and a significantly harder integration problem with no meaningful gain in output quality at the scale this prototype operates at.

Second, modern large language models are capable of performing both tasks through structured prompting alone. Using one model for everything keeps the pipeline clean, the outputs consistent, and the system explainable from one end to the other. Every output in the system comes from the same model following the same structured prompt logic, which makes it far easier to audit, trace, and defend. That transparency is not just good engineering practice; it is a direct mandate of the EU AI Act for high-risk AI systems.

4.2 Why Gemini 2.5 flash

Gemini 2.5 Flash is a large language model developed by Google DeepMind. It is built on transformer architecture, which is a deep learning framework that enables the model to understand context across long and complex inputs by learning the relationships between words, values, and patterns in data. That capability is central to this system because the model needs to read a row of numerical HR metrics and a block of free-text survey responses and reason meaningfully across both in the same call.

Four specific factors drove the selection of Gemini 2.5 Flash over alternative models.

The first is its context window. Gemini 2.5 Flash supports one of the largest context windows of any commercially available model, meaning it can process significantly more data in a single call than most alternatives. For a system handling workforce datasets with multiple data points per employee, this means the model can reason across the full dataset in one prompt rather than processing it in fragments, which produces more consistent and coherent outputs across the board.

The second is its performance on structured prompts. Every prompt in this system is precisely engineered to extract a specific output, whether that is a risk level, a sentiment classification, or a plain language explanation for a particular user role. Gemini 2.5 Flash has strong documented performance on instruction-following tasks, meaning it reliably produces the expected output format rather than drifting from the prompt structure. For a system where output consistency directly affects trust and auditability, reliability is a must.

The third is reasoning capability. Burnout signals are rarely straightforward. An employee with high overtime and zero leave days taken might be deeply engaged or on the edge of collapse, depending on a range of contextual factors. A model that simply pattern matches on surface-level thresholds will miss that nuance. Gemini 2.5 Flash is capable of weighing multiple signals simultaneously and producing a reasoned assessment rather than a mechanical one, which is what this use case demands.

The fourth is accessibility. Gemini 2.5 Flash is available through the Google AI Studio API, which provides a free tier that requires no payment credentials. At the data volumes involved in a prototype running on synthetic data, the free tier is more than sufficient. The API also integrates cleanly with Python through the Google-GenAI library and works directly within Google Colab, which is the development environment used for this prototype. That compatibility removes unnecessary technical friction and makes the system easier to build, test, and iterate on.

4.3 How the model is used in this system

Gemini 2.5 Flash is called at two distinct points in the pipeline.

The first call happens in the processing layer. The anonymised employee data is passed to the model alongside a structured prompt that instructs it to identify burnout risk signals in the numerical data, assess the sentiment of any free text survey responses, and return a structured output containing a risk level, the key signals that drove that assessment, and a confidence indicator. The prompt is deliberately constrained to the data provided, meaning the model is instructed to base its assessment only on what it has been given rather than drawing on generalised assumptions. This is what makes the output explainable and traceable rather than a black box verdict.

The second call happens in the generative AI layer. The combined risk profile produced by the first call is passed back to the model with a second structured prompt that instructs it to generate a plain language report tailored to the specific user role receiving it. The People Partner receives a full analytical report covering individual and team-level patterns. The Team Lead receives an aggregated team summary with no individual-level detail included in the prompt. The employee receives a personal well-being summary written in plain, simple language that explains what the system found and what it recommends.

Every prompt is version-controlled. That means every output can be traced back to the exact prompt and the exact data that produced it, which satisfies the EU AI Act requirement that automated decisions affecting individuals must be explainable and open to challenge.

4.4 Limitations of this model choice

Gemini 2.5 Flash is a general purpose language model. It was not specifically trained on HR data or validated against clinical burnout frameworks. Therefore, its understanding of burnout signals comes from broad pretraining rather than domain expertise, which means the quality of its assessments depends heavily on how well the prompts are engineered to direct its reasoning. This is a known limitation and is managed through structured prompt design, but it cannot be fully eliminated in a prototype at this stage.

The model also carries a hallucination risk. It can produce confident, well-written outputs that do not accurately reflect the underlying data. This is reduced by grounding every prompt in the actual data values before the model generates its summary, but human review of all outputs before any organisational action is taken remains a non-negotiable requirement of this system rather than an optional safeguard.

Gemini 2.5 Flash is also a closed-source commercial model. Its internal workings are not publicly available, which limits how deeply the system can be audited at the model level. This is why the prompt layer, the output structure, and the audit logging are all designed to be as transparent as possible within that constraint. The system cannot make the model fully transparent, but it can make everything around the model fully transparent, and that is the design principle on which this prototype is built.

5. Data Sources and Knowledge Retrieval

This system does not connect to any live organisational database or external HR platform. All data used in the development and testing of this prototype is synthetic, meaning it was artificially generated to reflect realistic HR patterns without containing any real employee information. This is a deliberate and justified design decision rather than a limitation of ambition.

5.1 Why synthetic data

Accessing real employee data for a project raises significant ethical and legal barriers. Employee records contain sensitive personal information protected under frameworks like GDPR, and obtaining legitimate access to that data outside of an institutional research agreement is not feasible at this stage. Using real data without proper consent and governance would directly contradict the privacy principles this system is designed to uphold, which would undermine the entire argument of the project.

Synthetic data solves this cleanly. It is generated to mirror the statistical patterns and distributions you would expect to find in a real workforce dataset, including realistic variation across risk levels, demographic groups, and behavioural signals, without any of the ethical or legal exposure that comes with real personal data. For a prototype whose purpose is to demonstrate that the system works as designed, synthetic data is not a compromise. It is the appropriate choice.

5.2 What the dataset contains

The synthetic dataset is structured as a CSV file with one row per employee. Each row contains two types of data that feed the two processing streams described in the system architecture.

The first type is quantitative. These are the numerical signals the system uses to assess burnout risk patterns. The dataset includes employee ID as a pseudonymous identifier, overtime hours logged per month, leave days taken in the past quarter, participation rate in team activities and meetings, output consistency score reflecting whether performance has been stable or declining, and a tenure value representing how long the employee has been in their current role. These fields were selected because they represent the most commonly cited behavioural indicators of burnout risk in organisational psychology research, specifically declining engagement, overwork, and withdrawal from team participation.

The second type is qualitative. Each row includes a synthetic survey response, a short piece of free text written to reflect the emotional tone and language patterns associated with different well-being states. Some responses reflect high stress and disengagement. Some reflect neutral or stable well-being. Some reflect positive engagement. This variation is intentional because the sentiment analysis prompt needs to be tested across a realistic range of inputs, not just worst-case scenarios.

5.3 How the data was generated

The dataset was generated with direct assistance from Claude, an AI assistant developed by Anthropic, using a structured generation approach that ensured realistic variation across risk profiles. The generation process created three distinct employee profiles: high risk, medium risk, and low risk, and populated each with numerical values and survey text that are internally consistent with that risk level. A high-risk employee row contains elevated overtime hours, zero leave days, low participation, declining output, and a survey response reflecting exhaustion and overwhelm. A medium-risk row reflects some warning signs alongside stable indicators and a survey response with cautious or uncertain language. A low-risk row reflects balanced working patterns and a survey response with positive or neutral language.

The dataset contains fifty rows distributed across the three risk categories, which is sufficient to test the full pipeline, evaluate prompt performance across different input types, and demonstrate the system working as intended in a prototype context.

5.4 Knowledge retrieval

In this system, knowledge retrieval does not operate as a separate technical component. There is no vector database, no document store, and no retrieval augmented generation pipeline in the traditional sense. Instead, the retrieval mechanism is the prompt itself.

Each time the system calls Gemini, the relevant employee data is retrieved from the dataset, formatted into a structured prompt, and passed directly to the model as context. The model reasons over that context in real time to produce its output. This approach is called in-context learning, meaning the model's knowledge about a specific employee comes entirely from what is provided in the prompt rather than from anything stored in an external knowledge base.

This is a deliberate and appropriate design choice for a prototype at this scale. In-context learning is sufficient when the data volume per call is manageable, which it is here, given Gemini 2.5 Flash's large context window. It also keeps the system architecture simple, transparent, and easy to audit, which aligns directly with the explainability requirements that run through this entire document.

A production version of this system, handling thousands of employees in real time, would benefit from a more sophisticated retrieval architecture, such as a vector database that stores employee profiles and dynamically retrieves the most relevant records. That is acknowledged as a natural evolution of the prototype rather than a flaw in the current design.

6. Deployment Infrastructure

This system is built as a working prototype using three tools: Google Colab, Streamlit, and the Gemini API. Each one was chosen for a specific reason, and together they form a deployment stack that is practical, accessible, and appropriate for a student capstone project operating within free-tier resource constraints.

6.1 Google Colab

Google Colab is the development environment where the entire AI pipeline is built and tested. It is a cloud-based Python notebook that runs in the browser, requires no local installation, and connects directly to Google Drive for persistent storage. For this project, it means the pipeline can be developed, tested, and demonstrated from any device without setting up a local development environment.

All the core logic lives here. The data ingestion, the anonymisation function, the assess_burnout_risk function, and the results collection all run inside Colab notebooks before being moved into the final Streamlit application. Colab also handles the connection to Google Drive, where the audit log is stored, ensuring that every action the system takes is recorded and persists across sessions.

The main limitation of Colab as a deployment environment is that it operates on session-based compute. When a session ends, the runtime resets and all variables are cleared. This means setup cells need to be rerun at the start of each session, which is manageable for a prototype but would not be acceptable in a production environment. This is acknowledged as a boundary of the prototype rather than a flaw in the system design.

6.2 Streamlit

Streamlit is the frontend framework used to turn the Python pipeline into an interactive web application. It takes the working Colab pipeline and wraps it in a browser-based interface that a non-technical user can actually use without touching any code. For this project, that means the People Partner can upload a CSV, the Team Lead can view a dashboard, the employee can access their personal summary, and the Compliance Officer can review audit logs, all through a clean interface built entirely in Python.

Streamlit was chosen over alternatives like Flask or Django for three reasons. Firstly, it requires significantly less code to produce a working interface, which is appropriate for a prototype at this scale. Secondly, it is designed specifically for data and AI applications, meaning components like file uploaders, data tables, and charts are built in rather than requiring additional libraries. Thirdly, it offers a free deployment tier on Streamlit Cloud, so the prototype can be shared via a public URL without any infrastructure costs or configuration.

The role-based access control described in the system architecture is implemented in Streamlit through a login selector that assigns each user to a role on sign-in. The application logic then filters what each role can see based on hardcoded permission rules. In a production system, this would be replaced by a proper authentication layer, but for a prototype demonstration, it is sufficient to demonstrate that the access control principle is working as intended. The prototype is deployed via Replit, a cloud-based development and hosting platform, which provides a stable public URL for demonstration purposes without requiring local Python installation or additional infrastructure configuration.

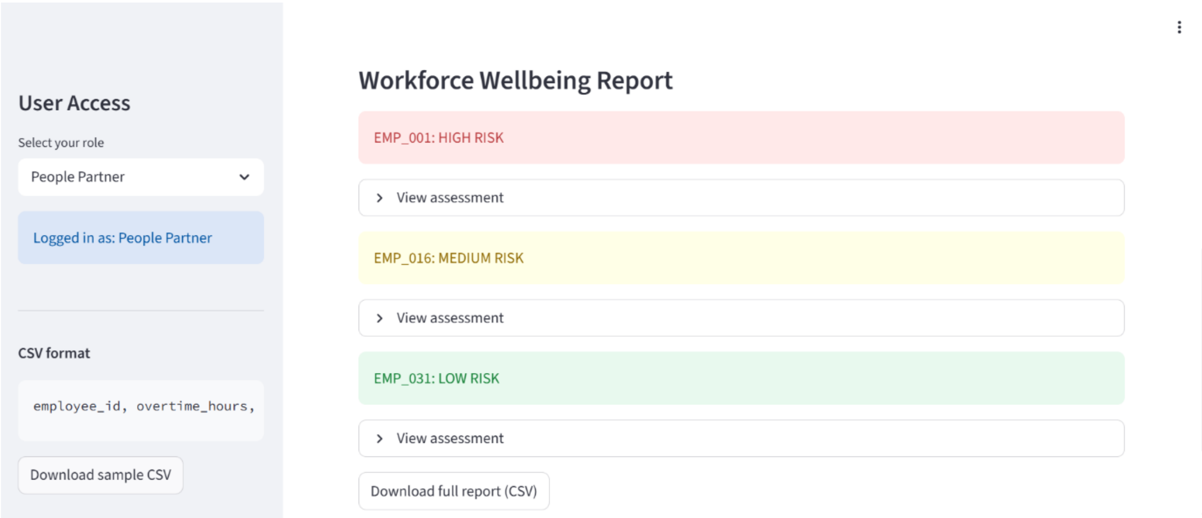

Figure 2: *People Partner Risk Report View.*

Figure 3: *Team Lead Aggregated Dashboard.*

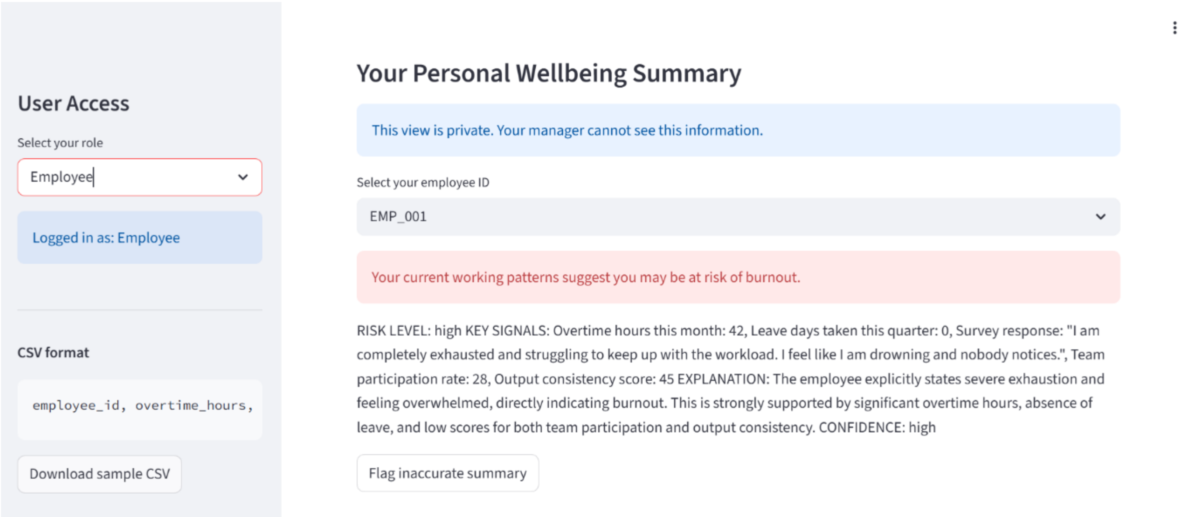

Figure 4: *Employee Personal Wellbeing Summary.*

Figure 5: *Compliance Officer Audit Dashboard.*

6.3 Gemini API

The Gemini API is the AI layer of the system. The prototype uses Gemini 2.5 Flash accessed through the Google AI Studio free tier via the Google Genai Python library. The model is called at two points in the pipeline, once to generate burnout risk assessments and once to generate role-specific plain language reports.

The free tier imposes rate limits on API calls, specifically a maximum number of requests per minute and a daily request quota. For a prototype running on a small synthetic dataset, this is manageable with deliberate spacing between calls using time.sleep in the pipeline code. At production scale with hundreds or thousands of employees, these limits would require either a paid API tier or a batch processing architecture that spaces calls across multiple sessions.

This is an acknowledged constraint of the prototype and is documented honestly rather than obscured. The rate-limiting issue encountered during development is itself a useful finding that informs the known limitations section and demonstrates a practical understanding of real-world API constraints that any production deployment of this system would need to address.

6.4 From prototype to production

The current deployment stack is appropriate for demonstrating that the system works as designed. A production version of this system would require a more robust infrastructure. The Colab notebook would be replaced by a dedicated backend server. The Streamlit interface would be replaced by or supplemented with a properly authenticated web application with role-based login tied to an identity provider. The free-tier Gemini API would be replaced by a paid tier with higher rate limits and SLA guarantees. Google Drive storage would be replaced by a proper database with encryption at rest and controlled access.

These are not design changes. The architecture, the prompt logic, the access control model, and the ethical principles embedded in the system all carry over directly. The production version is the same system running on infrastructure that can support real organisational use rather than a prototype demonstration.

7. Ethical and Security Considerations

This section does not treat ethics as an afterthought. The entire argument of this project is that HR AI systems have failed because they were built around efficiency rather than people. A system that claims to fix that problem while introducing its own ethical failures would be a contradiction. Every design decision documented in this section exists because of that starting point.

7.1 The surveillance paradox

The most significant ethical tension in this system is one that cannot be fully resolved, only honestly managed. Any system capable of detecting burnout early needs detailed enough data about how employees are working to do so meaningfully. That same level of detail is sufficient to constitute surveillance. The line between a system that protects employees and one that monitors them is not always clear, and it depends entirely on how the system is designed, who controls it, and whether employees trust it.

This prototype addresses that tension through three design decisions. First, individual employee data is never accessible to managers. That boundary is enforced at the architecture level through role-based access control, not just stated in a policy document. Second, every employee has access to a transparency interface that shows exactly what data the system holds on them and what it has concluded about them. Nothing about them is hidden from them. Third, the system is built on a consent-first principle, meaning employees are active participants with agency over their data rather than passive subjects being assessed without their knowledge.

These decisions reduce the surveillance risk but do not eliminate it entirely. A consent-first design only works if the consent is genuinely free. In a workplace context where employees may feel implicit pressure to participate, the line between voluntary consent and coerced consent is not always clear. This is an unresolved tension that any real-world deployment of this system would need to address through organisational policy and cultural change alongside the technical design.

7.2 Bias and algorithmic fairness

This system uses an LLM trained on broad internet data to assess burnout risk in employees. That model was not trained on HR data specifically, was not validated against clinical burnout frameworks, and reflects whatever biases exist in its training data. That is a real limitation, and it needs to be stated plainly.

The structured prompt design mitigates this by grounding every assessment in the specific data provided rather than allowing the model to draw on generalised assumptions. But mitigation is not elimination. If the prompts consistently produce outputs that disadvantage a particular group, whether by gender, age, ethnicity, or any other characteristic, the system will reproduce that bias at scale across every employee it assesses.

The Compliance Officer role exists precisely because of this risk. Their job is to continuously monitor system outputs for disproportionate patterns and investigate when they appear. But human oversight is only effective if the person doing the oversight has both the access and the mandate to act on what they find. Building the role into the system design is necessary but not sufficient. Organisational commitment to acting on biased findings is what makes it meaningful.

For a prototype using synthetic data, this risk is contained. The synthetic dataset was deliberately constructed with balanced representation across risk categories. In a real deployment with real organisational data, a bias audit before launch and continuous monitoring after would be non-negotiable requirements rather than recommendations.

7.3 Data privacy and security

This system processes sensitive personal data. Attendance patterns, survey responses, and wellbeing assessments are not neutral information. In the wrong hands or used for the wrong purposes they could damage careers, expose vulnerabilities, or be used to justify decisions that have nothing to do with employee wellbeing.

The prototype addresses this through pseudonymisation at the point of ingestion. Employee names and identifiers are stripped and replaced with pseudonymous IDs before any data reaches the LLM. The mapping table that links those IDs back to real identities is stored separately and is only accessible to the system administrator. The LLM never processes a real name.

Two limitations are worth being honest about here. First, pseudonymisation is not the same as full anonymisation. With enough contextual information it is theoretically possible to re-identify an individual from their data even without a name attached. In a small team, this risk is higher. Second, the Gemini API is a third-party commercial service. When data is sent to the API, it leaves the local environment and is processed on Google's servers. For a prototype using synthetic data, this is acceptable. For a production system handling real employee data, it would require a data processing agreement with Google and a legal review of whether that arrangement complies with GDPR and any applicable employment law.

7.4 Explainability and the right to contest

The EU AI Act classifies consequential HR AI tools as high-risk systems. One of the core requirements for high-risk systems is that individuals affected by automated decisions must be able to understand how those decisions were made and have a meaningful route to contest them.

This system addresses that requirement in two ways. Every LLM output includes an explanation of which signals drove the assessment, not just a risk label. And every employee has access to a feedback mechanism that lets them flag if they believe the system's output does not reflect their experience. That feedback is routed into the bias monitoring process rather than disappearing into a void.

The honest limitation here is that the LLM explanation layer carries a hallucination risk. The model can produce a coherent and confident explanation that does not accurately reflect the underlying reasoning. This is reduced by grounding prompts tightly in the actual data, but it cannot be eliminated entirely. Human review of all outputs before any action is taken is built into the system as a requirement, not a suggestion. The system is designed to inform human decisions, not replace them.

7.5 Duty of care and organisational responsibility

This system gives organisations tools they did not previously have. The People Partner can now see early warning signals that would have been invisible before. The Team Lead can identify patterns of strain before they become crises. That is the point of the system, and it is genuinely valuable.

But access to that information creates responsibility. An organisation that uses this system and then ignores what it surfaces has not improved its duty of care; it has just documented its failure to act. The system does not make organisations ethical. It gives ethical organisations better tools to act on their values and gives unethical ones more data to ignore or misuse.

This is the hardest limitation to design around because it is not a technical problem. It is a human and organisational one. The most honest thing this document can say is that the system is a tool, and tools are only as good as the intent of the people using them. Everything in this design attempts to make misuse harder and ethical use easier. Whether that is enough depends on the organisation deploying it.

References

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